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Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

*Equal contribution. Listing order is random. Jakob proposed replacing RNNs with self-attention and started the effort to evaluate this idea. Ashish, with Illia, designed and implemented the first Transformer models and has been crucially involved in every aspect of this work. Noam proposed scaled dot-product attention, multi-head attention and the parameter-free position representation and became the other person involved in nearly every detail. Niki designed, implemented, tuned and evaluated countless model variants in our original codebase and tensor2tensor. Llion also experimented with novel model variants, was responsible for our initial codebase, and efficient inference and visualizations. Lukasz and Aidan spent countless long days designing various parts of and implementing tensor2tensor, replacing our earlier codebase, greatly improving results and massively accelerating our research.

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注意力就是你所需要的一切

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摘要

主流的序列转导模型基于复杂的循环或卷积神经网络，这些网络包含编码器和解码器。表现最好的模型还通过注意力机制连接编码器和解码器。我们提出了一种新的简单网络架构——Transformer，它完全基于注意力机制，彻底摒弃了循环和卷积。在两个机器翻译任务上的实验表明，这些模型质量更优，同时更易于并行化，所需训练时间也显著减少。我们的模型在 WMT 2014 英德翻译任务上取得了 28.4 BLEU，比现有最佳结果（包括集成模型）高出 2 BLEU 以上。在 WMT 2014 英法翻译任务上，我们的模型在 8 个 GPU 上训练 3.5 天后，建立了新的单模型最先进 BLEU 分数 41.8，而训练成本仅为文献中最佳模型的一小部分。我们通过将 Transformer 成功应用于英语成分句法分析（无论训练数据充足还是有限），表明它能很好地泛化到其他任务。

*同等贡献。作者排序为随机。Jakob 提出用自注意力替代 RNN，并启动了评估这一想法的努力。Ashish 与 Illia 共同设计并实现了最初的 Transformer 模型，并在本项工作的各个方面发挥了关键作用。Noam 提出了缩放点积注意力、多头注意力和无参数位置表示，并成为几乎参与每个细节的另一人。Niki 在我们最初的代码库和 tensor2tensor 中设计、实现、调优并评估了无数模型变体。Llion 还试验了新颖的模型变体，负责我们最初的代码库以及高效推理和可视化。Lukasz 和 Aidan 花费无数漫长时日设计并实现 tensor2tensor 的各个部分，替换了我们早先的代码库，极大提升了结果，并大幅加速了我们的研究。

[†]本工作是在 Google Brain 期间完成的。

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1 Introduction

Recurrent neural networks, long short-term memory [13] and gated recurrent [7] neural networks in particular, have been firmly established as state of the art approaches in sequence modeling and transduction problems such as language modeling and machine translation [35, 2, 5]. Numerous efforts have since continued to push the boundaries of recurrent language models and encoder-decoder architectures [38, 24, 15].

Recurrent models typically factor computation along the symbol positions of the input and output sequences. Aligning the positions to steps in computation time, they generate a sequence of hidden states h_t , as a function of the previous hidden state h_{t-1} and the input for position t . This inherently sequential nature precludes parallelization within training examples, which becomes critical at longer sequence lengths, as memory constraints limit batching across examples. Recent work has achieved significant improvements in computational efficiency through factorization tricks [21] and conditional computation [32], while also improving model performance in case of the latter. The fundamental constraint of sequential computation, however, remains.

Attention mechanisms have become an integral part of compelling sequence modeling and transduction models in various tasks, allowing modeling of dependencies without regard to their distance in the input or output sequences [2, 19]. In all but a few cases [27], however, such attention mechanisms are used in conjunction with a recurrent network.

In this work we propose the Transformer, a model architecture eschewing recurrence and instead relying entirely on an attention mechanism to draw global dependencies between input and output. The Transformer allows for significantly more parallelization and can reach a new state of the art in translation quality after being trained for as little as twelve hours on eight P100 GPUs.

2 Background

The goal of reducing sequential computation also forms the foundation of the Extended Neural GPU [16], ByteNet [18] and ConvS2S [9], all of which use convolutional neural networks as basic building block, computing hidden representations in parallel for all input and output positions. In these models, the number of operations required to relate signals from two arbitrary input or output positions grows in the distance between positions, linearly for ConvS2S and logarithmically for ByteNet. This makes it more difficult to learn dependencies between distant positions [12]. In the Transformer this is reduced to a constant number of operations, albeit at the cost of reduced effective resolution due to averaging attention-weighted positions, an effect we counteract with Multi-Head Attention as described in section 3.2.

Self-attention, sometimes called intra-attention is an attention mechanism relating different positions of a single sequence in order to compute a representation of the sequence. Self-attention has been used successfully in a variety of tasks including reading comprehension, abstractive summarization, textual entailment and learning task-independent sentence representations [4, 27, 28, 22].

End-to-end memory networks are based on a recurrent attention mechanism instead of sequence-aligned recurrence and have been shown to perform well on simple-language question answering and language modeling tasks [34].

To the best of our knowledge, however, the Transformer is the first transduction model relying entirely on self-attention to compute representations of its input and output without using sequence-aligned RNNs or convolution. In the following sections, we will describe the Transformer, motivate self-attention and discuss its advantages over models such as [17, 18] and [9].

3 Model Architecture

Most competitive neural sequence transduction models have an encoder-decoder structure [5, 2, 35]. Here, the encoder maps an input sequence of symbol representations $(x_1, \dots, x_n)$ to a sequence of continuous representations $\mathbf{z} = (z_1, \dots, z_n)$. Given $\mathbf{z}$, the decoder then generates an output sequence $(y_1, \dots, y_m)$ of symbols one element at a time. At each step the model is auto-regressive [10], consuming the previously generated symbols as additional input when generating the next.

1 引言

循环神经网络，尤其是长短期记忆 [13] 和门控循环 [7] 神经网络，已被牢固确立为序列建模和转导问题（如语言建模和机器翻译）中的最先进方法 [35, 2, 5]。此后，大量努力不断拓展循环语言模型和编码器-解码器架构的边界 [38, 24, 15]。

循环模型通常按照输入和输出序列的符号位置来分解计算。将位置与计算时间步对齐，它们会生成隐藏状态序列 h_t ，该序列是前一隐藏状态 h_{t-1} 和位置 t 的输入的函数。这种固有的序列性阻碍了训练样本内的并行化；当序列长度较长时，这一点变得尤为关键，因为内存限制限制了跨样本的批处理。近期工作通过因式分解技巧 [21] 和条件计算 [32]，在计算效率上取得了显著提升，同时在后者的情况下也提高了模型性能。然而，顺序计算的约束依然存在。

注意力机制已成为各种任务中引人注目的序列建模和转导模型不可或缺的一部分，使得能够对输入或输出序列 [2, 19] 中的依赖关系进行建模，而无需考虑其距离。然而，在除少数情况 [27]，之外的所有情况下，此类注意力机制都与循环网络结合使用。

在这项工作中，我们提出了 Transformer，这是一种摒弃循环、转而完全依赖注意力机制来提取输入与输出之间全局依赖关系的模型架构。Transformer 允许显著更多的并行化，并且在八块 P100 GPU 上仅训练十二小时后，就能在翻译质量上达到新的最先进水平。

2 背景

减少顺序计算的目标也构成了扩展神经GPU[16]、ByteNet [18] 和 ConvS2S [9] 的基础，它们都使用卷积神经网络作为基本构建模块，并为所有输入和输出位置并行计算隐藏表示。在这些模型中，将来自任意两个输入或输出位置的信号关联起来所需的操作数随位置之间的距离而增长：对于 ConvS2S 是线性的，对于 ByteNet 是对数级的。这使得学习远距离位置之间的依赖关系变得更加困难 [12]。在 Transformer 中，这被减少到常数次操作，尽管代价是由于对注意力加权位置进行平均而导致有效分辨率降低；我们通过第 3.2 节所述的多头注意力来抵消这一影响。

自注意力，有时也称为内部注意力，是一种注意力机制，它将单个序列的不同位置关联起来，以计算该序列的表示。自注意力已成功应用于多种任务，包括阅读理解、抽象式摘要、文本蕴含以及学习与任务无关的句子表示 [4, 27, 28, 22]。

端到端记忆网络基于循环注意力机制，而不是序列对齐的循环，并且已被证明在简单语言问答和语言任务 [34] 上表现良好。

然而，据我们所知，Transformer 是第一个完全依赖自注意力来计算其输入和输出表示的转导模型，而无需使用序列对齐的 RNN 或。在接下来的章节中，我们将描述 Transformer，说明自注意力的动机，并讨论其相较于 [17, 18] 和 [9] 等模型的优势。

3 模型架构

大多数具有竞争力的神经序列转导模型都采用编码器-解码器结构 [5, 2, 35]。在这里，编码器将符号表示的输入序列 $(x_1, \dots, x_n)$ 映射到连续表示序列 $\mathbf{z} = (z_1, \dots, z_n)$ 。给定 $\mathbf{z}$ ，解码器随后一次一个元素地生成由符号组成的输出序列 $(y_1, \dots, y_m)$ 。在每一步，模型都是自回归的 [10]，在生成下一个符号时利用先前生成的符号作为额外输入。

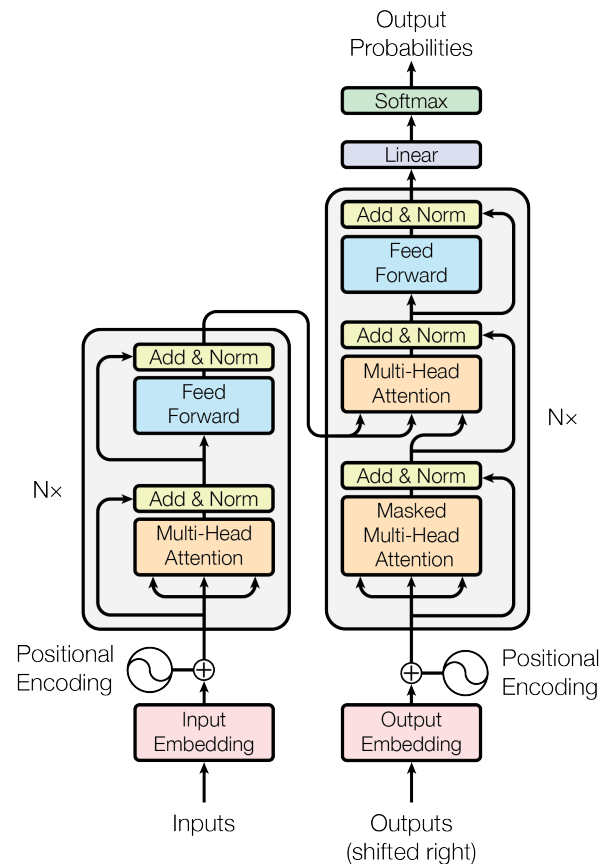


Figure 1: The Transformer - model architecture.

The Transformer follows this overall architecture using stacked self-attention and point-wise, fully connected layers for both the encoder and decoder, shown in the left and right halves of Figure 1, respectively.

3.1 Encoder and Decoder Stacks

Encoder: The encoder is composed of a stack of $N = 6$ identical layers. Each layer has two sub-layers. The first is a multi-head self-attention mechanism, and the second is a simple, position-wise fully connected feed-forward network. We employ a residual connection [11] around each of the two sub-layers, followed by layer normalization [1]. That is, the output of each sub-layer is $\text{LayerNorm}(x + \text{Sublayer}(x))$, where $\text{Sublayer}(x)$ is the function implemented by the sub-layer itself. To facilitate these residual connections, all sub-layers in the model, as well as the embedding layers, produce outputs of dimension $d_{\text{model}} = 512$.

Decoder: The decoder is also composed of a stack of $N = 6$ identical layers. In addition to the two sub-layers in each encoder layer, the decoder inserts a third sub-layer, which performs multi-head attention over the output of the encoder stack. Similar to the encoder, we employ residual connections around each of the sub-layers, followed by layer normalization. We also modify the self-attention sub-layer in the decoder stack to prevent positions from attending to subsequent positions. This masking, combined with fact that the output embeddings are offset by one position, ensures that the predictions for position i can depend only on the known outputs at positions less than i .

3.2 Attention

An attention function can be described as mapping a query and a set of key-value pairs to an output, where the query, keys, values, and output are all vectors. The output is computed as a weighted sum

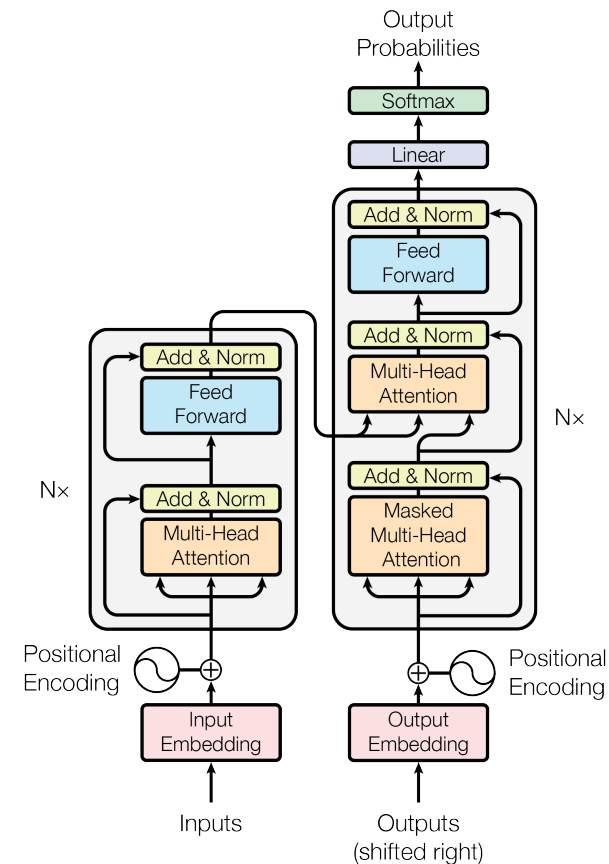


图 1: Transformer - 模型架构。

Transformer 遵循这一整体架构，对编码器和解码器都使用堆叠式自注意力和逐位置全连接层，分别如图 1 的左半部分和右半部分所示。

3.1 编码器和解码器堆栈

编码器：编码器由 $N = 6$ 个相同的层堆叠而成。每一层都有两个子层。第一个是多头自注意力机制，第二个是简单的逐位置全连接前馈网络。我们在两个子层中的每一个周围使用残差连接 [11]，随后进行层归一化 [1]。也就是说，每个子层的输出为 $\text{LayerNorm}(x + \text{Sublayer}(x))$ ，其中 $\text{Sublayer}(x)$ 是该子层自身实现的函数。为了便于实现这些残差连接，模型中的所有子层以及嵌入层都产生维度为 $d_{\text{model}} = 512$ 的输出。

解码器：解码器同样由 $N = 6$ 个相同的层堆叠而成。除了每个编码器层中的两个子层外，解码器还插入了第三个子层，该子层对编码器堆叠的输出执行多头注意力。与编码器类似，我们在每个子层周围使用残差连接，随后进行层归一化。我们还修改了解码器堆叠中的自注意力子层，以防止位置关注后续位置。这种掩码与输出嵌入偏移一个位置这一事实相结合，确保了对位置 i 的预测只能依赖于位置小于 i 的已知输出。

3.2 注意力

注意力函数可以描述为将一个查询和一组键值对映射为一个输出，其中查询、键、值和输出都是向量。输出被计算为加权和



Figure 2: (left) Scaled Dot-Product Attention. (right) Multi-Head Attention consists of several attention layers running in parallel.

of the values, where the weight assigned to each value is computed by a compatibility function of the query with the corresponding key.

3.2.1 Scaled Dot-Product Attention

We call our particular attention "Scaled Dot-Product Attention" (Figure 2). The input consists of queries and keys of dimension d_k , and values of dimension d_v . We compute the dot products of the query with all keys, divide each by $\sqrt{d_k}$, and apply a softmax function to obtain the weights on the values.

In practice, we compute the attention function on a set of queries simultaneously, packed together into a matrix Q . The keys and values are also packed together into matrices K and V . We compute the matrix of outputs as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

The two most commonly used attention functions are additive attention [2], and dot-product (multiplicative) attention. Dot-product attention is identical to our algorithm, except for the scaling factor of $\frac{1}{\sqrt{d_k}}$. Additive attention computes the compatibility function using a feed-forward network with a single hidden layer. While the two are similar in theoretical complexity, dot-product attention is much faster and more space-efficient in practice, since it can be implemented using highly optimized matrix multiplication code.

While for small values of d_k the two mechanisms perform similarly, additive attention outperforms dot product attention without scaling for larger values of d_k [3]. We suspect that for large values of d_k , the dot products grow large in magnitude, pushing the softmax function into regions where it has extremely small gradients⁴. To counteract this effect, we scale the dot products by $\frac{1}{\sqrt{d_k}}$.

3.2.2 Multi-Head Attention

Instead of performing a single attention function with d_{model} -dimensional keys, values and queries, we found it beneficial to linearly project the queries, keys and values h times with different, learned linear projections to d_k , d_k and d_v dimensions, respectively. On each of these projected versions of queries, keys and values we then perform the attention function in parallel, yielding d_v -dimensional

⁴To illustrate why the dot products get large, assume that the components of q and k are independent random variables with mean 0 and variance 1. Then their dot product, $q \cdot k = \sum_{i=1}^{d_k} q_i k_i$, has mean 0 and variance d_k .

图 2：（左）缩放点积注意力。（右）多头注意力由多个并行运行的注意力层组成。

这些值的加权和，其中分配给每个值的权重由查询与对应键的兼容性函数计算得出。

3.2.1 缩放点积注意力

我们将这种特定的注意力称为“缩放点积注意力”（图 2）。输入由维度为 d_k 的查询和键，以及维度为 d_v 的值组成。我们计算查询与所有键的点积，将每个点积除以 $\sqrt{d_k}$ ，并应用 softmax 函数以获得作用于值的权重。

在实践中，我们同时在一组查询上计算注意力函数，将它们打包成一个矩阵 Q 。键和值也分别打包成矩阵 K 和 V 。我们将输出矩阵计算为：

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

两种最常用的注意力函数是加性注意力 [2]，和点积（乘性）注意力。点积注意力与我们的算法相同，只是缩放因子为 $\frac{1}{\sqrt{d_k}}$ 。加性注意力使用具有单个隐藏层的前馈网络计算兼容性函数。尽管两者在理论复杂度上相似，但在实践中，点积注意力要快得多，也更节省空间，因为它可以使用高度优化的矩阵乘法代码来实现。

当 d_k 的值较小时，这两种机制表现相似；而对于较大的 d_k [3] 值，加性注意力优于未进行缩放的点积注意力。我们推测，对于较大的 d_k 值，点积的幅值会变得很大，从而将 softmax 函数推入梯度极小的区域⁴。为抵消这种影响，我们按 $\frac{1}{\sqrt{d_k}}$ 缩放点积。

3.2.2 多头注意力

与其对 d_{model} 维的键、值和查询执行单一的注意力函数，我们发现，用不同的、学习到的线性投影将查询、键和值分别线性投影 h 次到 d_k 、 d_k 和 d_v 维会更有意义。随后，在查询、键和值的每个投影版本上，我们并行执行注意力函数，得到 d_v 维的

⁴为了说明点积为何会变大，假设 q 和 k 的分量是均值为 0、方差为 1 的独立随机变量。那么它们的点积 $q \cdot k = \sum_{i=1}^{d_k} q_i k_i$ 的均值为 0，方差为 d_k 。

output values. These are concatenated and once again projected, resulting in the final values, as depicted in Figure 2.

Multi-head attention allows the model to jointly attend to information from different representation subspaces at different positions. With a single attention head, averaging inhibits this.

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O$$

$$\text{where head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$

Where the projections are parameter matrices $W_i^Q \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_i^K \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_i^V \in \mathbb{R}^{d_{\text{model}} \times d_v}$ and $W^O \in \mathbb{R}^{hd_v \times d_{\text{model}}}$.

In this work we employ $h = 8$ parallel attention layers, or heads. For each of these we use $d_k = d_v = d_{\text{model}}/h = 64$. Due to the reduced dimension of each head, the total computational cost is similar to that of single-head attention with full dimensionality.

3.2.3 Applications of Attention in our Model

The Transformer uses multi-head attention in three different ways:

- In "encoder-decoder attention" layers, the queries come from the previous decoder layer, and the memory keys and values come from the output of the encoder. This allows every position in the decoder to attend over all positions in the input sequence. This mimics the typical encoder-decoder attention mechanisms in sequence-to-sequence models such as [38, 2, 9].
- The encoder contains self-attention layers. In a self-attention layer all of the keys, values and queries come from the same place, in this case, the output of the previous layer in the encoder. Each position in the encoder can attend to all positions in the previous layer of the encoder.
- Similarly, self-attention layers in the decoder allow each position in the decoder to attend to all positions in the decoder up to and including that position. We need to prevent leftward information flow in the decoder to preserve the auto-regressive property. We implement this inside of scaled dot-product attention by masking out (setting to $-\infty$) all values in the input of the softmax which correspond to illegal connections. See Figure 2.

3.3 Position-wise Feed-Forward Networks

In addition to attention sub-layers, each of the layers in our encoder and decoder contains a fully connected feed-forward network, which is applied to each position separately and identically. This consists of two linear transformations with a ReLU activation in between.

$$\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2 \quad (2)$$

While the linear transformations are the same across different positions, they use different parameters from layer to layer. Another way of describing this is as two convolutions with kernel size 1. The dimensionality of input and output is $d_{\text{model}} = 512$, and the inner-layer has dimensionality $d_{ff} = 2048$.

3.4 Embeddings and Softmax

Similarly to other sequence transduction models, we use learned embeddings to convert the input tokens and output tokens to vectors of dimension d_{model} . We also use the usual learned linear transformation and softmax function to convert the decoder output to predicted next-token probabilities. In our model, we share the same weight matrix between the two embedding layers and the pre-softmax linear transformation, similar to [30]. In the embedding layers, we multiply those weights by $\sqrt{d_{\text{model}}}$.

输出值。这些输出值被拼接起来，并再次投影，得到最终值，如图 2 所示。

多头注意力允许模型同时关注来自不同表示子空间中不同位置的信息。使用单个注意力头时，平均会抑制这一点。

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O$$

$$\text{where head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$

其中，这些投影是参数矩阵 $W_i^Q \in \mathbb{R}^{d_{\text{model}} \times d_k}$ 、 $W_i^K \in \mathbb{R}^{d_{\text{model}} \times d_k}$ 、 $W_i^V \in \mathbb{R}^{d_{\text{model}} \times d_v}$ 和 $W^O \in \mathbb{R}^{hd_v \times d_{\text{model}}}$ 。

在这项工作中，我们采用 $h = 8$ 个并行注意力层，或称头。对于其中每一个，我们使用 $d_k = d_v = d_{\text{model}}/h = 64$ 。由于每个头的维度降低，总计算成本与具有完整维度的单头注意力相似。

3.2.3 注意力在我们的模型中的应用

Transformer 通过三种不同方式使用多头注意力：

- 在“编码器-解码器注意力”层中，查询来自前一个解码器层，而记忆的键和值来自编码器的输出。这使得解码器中的每个位置都能关注输入序列中的所有位置。这模仿了诸如[38, 2, 9]等序列到序列模型中的典型编码器-解码器注意力机制。
- 编码器包含自注意力层。在自注意力层中，所有键、值和查询都来自同一个地方，在此情况下即编码器中前一层的输出。编码器中的每个位置都能关注到编码器前一层的所有位置。
- 类似地，解码器中的自注意力层允许解码器中的每个位置关注解码器中直到并包括该位置在内的所有位置。我们需要防止解码器中的信息向左流动，以保持自回归特性。我们通过在缩放点积注意力中屏蔽掉（设置为 $-\infty$ ）softmax 输入中所有对应于非法连接的值来实现这一点。参见图 2。

3.3 逐位置前馈网络

除了注意力子层之外，我们的编码器和解码器中的每一层都包含一个全连接前馈网络，该网络分别且相同地应用于每个位置。它由两个线性变换组成，中间有一个 ReLU 激活函数。

$$\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2 \quad (2)$$

虽然线性变换在不同位置上是相同的，但它们在层与层之间使用不同的参数。另一种描述方式是将其视为两个卷积核大小为 1 的卷积。输入和输出的维度为 $d_{\text{model}} = 512$ ，而内层的维度为 $d_{ff} = 2048$ 。

3.4 嵌入与 Softmax

与其他序列转导模型类似，我们使用学习到的嵌入将输入词元和输出词元转换为维度为 d_{model} 的向量。我们还使用通常的学习线性变换和 softmax 函数，将解码器输出转换为预测的下一词元概率。在我们的模型中，我们在两个嵌入层和 softmax 前线性变换之间共享同一个权重矩阵，类似于 [30]。在嵌入层中，我们将这些权重乘以 $\sqrt{d_{\text{model}}}$ 。

Table 1: Maximum path lengths, per-layer complexity and minimum number of sequential operations for different layer types. n is the sequence length, d is the representation dimension, k is the kernel size of convolutions and r the size of the neighborhood in restricted self-attention.

Layer Type	Complexity per Layer	Sequential Operations	Maximum Path Length
Self-Attention	$O(n^2 \cdot d)$	$O(1)$	$O(1)$
Recurrent	$O(n \cdot d^2)$	$O(n)$	$O(n)$
Convolutional	$O(k \cdot n \cdot d^2)$	$O(1)$	$O(\log_k(n))$
Self-Attention (restricted)	$O(r \cdot n \cdot d)$	$O(1)$	$O(n/r)$

3.5 Positional Encoding

Since our model contains no recurrence and no convolution, in order for the model to make use of the order of the sequence, we must inject some information about the relative or absolute position of the tokens in the sequence. To this end, we add "positional encodings" to the input embeddings at the bottoms of the encoder and decoder stacks. The positional encodings have the same dimension d_{model} as the embeddings, so that the two can be summed. There are many choices of positional encodings, learned and fixed [9].

In this work, we use sine and cosine functions of different frequencies:

$$PE_{(pos, 2i)} = \sin(pos/10000^{2i/d_{\text{model}}})$$

$$PE_{(pos, 2i+1)} = \cos(pos/10000^{2i/d_{\text{model}}})$$

where pos is the position and i is the dimension. That is, each dimension of the positional encoding corresponds to a sinusoid. The wavelengths form a geometric progression from 2π to $10000 \cdot 2\pi$. We chose this function because we hypothesized it would allow the model to easily learn to attend by relative positions, since for any fixed offset k , PE_{pos+k} can be represented as a linear function of PE_{pos} .

We also experimented with using learned positional embeddings [9] instead, and found that the two versions produced nearly identical results (see Table 3 row (E)). We chose the sinusoidal version because it may allow the model to extrapolate to sequence lengths longer than the ones encountered during training.

4 Why Self-Attention

In this section we compare various aspects of self-attention layers to the recurrent and convolutional layers commonly used for mapping one variable-length sequence of symbol representations $(x_1, \dots, x_n)$ to another sequence of equal length $(z_1, \dots, z_n)$, with $x_i, z_i \in \mathbb{R}^d$, such as a hidden layer in a typical sequence transduction encoder or decoder. Motivating our use of self-attention we consider three desiderata.

One is the total computational complexity per layer. Another is the amount of computation that can be parallelized, as measured by the minimum number of sequential operations required.

The third is the path length between long-range dependencies in the network. Learning long-range dependencies is a key challenge in many sequence transduction tasks. One key factor affecting the ability to learn such dependencies is the length of the paths forward and backward signals have to traverse in the network. The shorter these paths between any combination of positions in the input and output sequences, the easier it is to learn long-range dependencies [12]. Hence we also compare the maximum path length between any two input and output positions in networks composed of the different layer types.

As noted in Table 1, a self-attention layer connects all positions with a constant number of sequentially executed operations, whereas a recurrent layer requires $O(n)$ sequential operations. In terms of computational complexity, self-attention layers are faster than recurrent layers when the sequence

表1: 不同层类型的最大路径长度、每层复杂度和最少顺序操作数。 n 是序列长度, d 是表示维度, k 是卷积核大小, r 是受限自注意力中邻域的大小。

层类型	每层复杂度	顺序操作	最大路径长度
自注意力	$O(n^2 \cdot d)$	$O(1)$	$O(1)$
循环	$O(n \cdot d^2)$	$O(n)$	$O(n)$
卷积	$O(k \cdot n \cdot d^2)$	$O(1)$	$O(\log_k(n))$
自注意力 (受限)	$O(r \cdot n \cdot d)$	$O(1)$	$O(n/r)$

3.5 位置编码

由于我们的模型既不含循环也不含卷积, 为了使模型能够利用序列的顺序, 我们必须注入一些关于序列中词元的相对或绝对位置的信息。为此, 我们在编码器和解码器堆栈底部的输入嵌入中加入“位置编码”。位置编码与嵌入具有相同的维度 d_{model} , 因此二者可以相加。位置编码有多种选择, 既有学习得到的, 也有固定的 [9]。

在这项工作中, 我们使用不同频率的正弦和余弦函数:

$$PE_{(pos, 2i)} = \sin(pos/10000^{2i/d_{\text{model}}})$$

$$PE_{(pos, 2i+1)} = \cos(pos/10000^{2i/d_{\text{model}}})$$

其中 pos 是位置, i 是维度。也就是说, 位置编码的每个维度都对应一个正弦波。波长从 2π 到 $10000 \cdot 2\pi$ 构成几何级数。我们选择这个函数, 是因为我们假设它能让模型轻松学会按相对位置进行注意力计算, 因为对于任何固定偏移 k , PE_{pos+k} 都可以表示为 PE_{pos} 的线性函数。

我们还实验了改用学习到的位置嵌入 [9], 并发现这两种版本产生的结果几乎相同 (见表 3 第 (E) 行)。我们选择正弦版本, 是因为它可能使模型能够外推到比训练期间遇到的序列更长的序列长度。

4 为什么使用自注意力

在本节中, 我们将自注意力层的各个方面与常用的循环层和卷积层进行比较, 这些层用于将一个变长符号表示序列 $(x_1, \dots, x_n)$ 映射到另一个等长序列 $(z_1, \dots, z_n)$, 其中 $x_i, z_i \in \mathbb{R}^d$, 例如典型序列转换编码器或解码器中的隐藏层。为了说明我们为何使用自注意力, 我们考虑三个期望特性。

一是每层的总计算复杂度。另一个是可并行化的计算量, 以所需的最少顺序操作数来衡量。

第三是网络中长距离依赖之间的路径长度。学习长距离依赖是许多序列转换任务中的关键挑战。影响学习此类依赖能力的一个关键因素是前向和后向信号在网络中必须穿行的路径长度。输入序列和输出序列中任意位置组合之间的这些路径越短, 就越容易学习长距离依赖 [12]。因此, 我们还在由不同层类型组成的网络中, 比较任意两个输入和输出位置之间的最大路径长度。

如表 1 所述, 自注意力层以常数个顺序执行的操作连接所有位置, 而循环层需要 $O(n)$ 个顺序操作。就计算复杂度而言, 当序列

length n is smaller than the representation dimensionality d , which is most often the case with sentence representations used by state-of-the-art models in machine translations, such as word-piece [38] and byte-pair [31] representations. To improve computational performance for tasks involving very long sequences, self-attention could be restricted to considering only a neighborhood of size r in the input sequence centered around the respective output position. This would increase the maximum path length to $O(n/r)$. We plan to investigate this approach further in future work.

A single convolutional layer with kernel width $k < n$ does not connect all pairs of input and output positions. Doing so requires a stack of $O(n/k)$ convolutional layers in the case of contiguous kernels, or $O(\log_k(n))$ in the case of dilated convolutions [18], increasing the length of the longest paths between any two positions in the network. Convolutional layers are generally more expensive than recurrent layers, by a factor of k . Separable convolutions [6], however, decrease the complexity considerably, to $O(k \cdot n \cdot d + n \cdot d^2)$. Even with $k = n$, however, the complexity of a separable convolution is equal to the combination of a self-attention layer and a point-wise feed-forward layer, the approach we take in our model.

As side benefit, self-attention could yield more interpretable models. We inspect attention distributions from our models and present and discuss examples in the appendix. Not only do individual attention heads clearly learn to perform different tasks, many appear to exhibit behavior related to the syntactic and semantic structure of the sentences.

5 Training

This section describes the training regime for our models.

5.1 Training Data and Batching

We trained on the standard WMT 2014 English-German dataset consisting of about 4.5 million sentence pairs. Sentences were encoded using byte-pair encoding [3], which has a shared source-target vocabulary of about 37000 tokens. For English-French, we used the significantly larger WMT 2014 English-French dataset consisting of 36M sentences and split tokens into a 32000 word-piece vocabulary [38]. Sentence pairs were batched together by approximate sequence length. Each training batch contained a set of sentence pairs containing approximately 25000 source tokens and 25000 target tokens.

5.2 Hardware and Schedule

We trained our models on one machine with 8 NVIDIA P100 GPUs. For our base models using the hyperparameters described throughout the paper, each training step took about 0.4 seconds. We trained the base models for a total of 100,000 steps or 12 hours. For our big models,(described on the bottom line of table 3), step time was 1.0 seconds. The big models were trained for 300,000 steps (3.5 days).

5.3 Optimizer

We used the Adam optimizer [20] with $\beta_1 = 0.9$, $\beta_2 = 0.98$ and $\epsilon = 10^{-9}$. We varied the learning rate over the course of training, according to the formula:

$$lrate = d_{\text{model}}^{-0.5} \cdot \min(step_num^{-0.5}, step_num \cdot warmup_steps^{-1.5}) \quad (3)$$

This corresponds to increasing the learning rate linearly for the first $warmup_steps$ training steps, and decreasing it thereafter proportionally to the inverse square root of the step number. We used $warmup_steps = 4000$.

5.4 Regularization

We employ three types of regularization during training:

长度 n 小于表示维度 d , 这在机器翻译中最先进模型所使用的句子表示中最为常见, 例如词片[38] 和字节对 [31] 表示。对于涉及很长序列的任务, 为提高计算性能, 可以将自注意力限制为仅考虑输入序列中以相应输出为中心、大小为 r 的邻域。这会将最大路径长度增加到 $O(n/r)$ 。我们计划在未来的工作中进一步研究这种方法。

核宽度为 $k < n$ 的单个卷积层并不能连接输入和输出的所有位置对。要连接所有位置对, 在连续卷积核的情况下需要堆叠 $O(n/k)$ 个卷积层, 而在空洞卷积的情况下需要 $O(\log_k(n))$ 个, [18], 这增加了网络中任意两个位置之间最长路径的长度。卷积层通常比循环层更昂贵, 其代价是 k 倍。然而, 可分离卷积 [6], 将复杂度显著降低到 $O(k \cdot n \cdot d + n \cdot d^2)$ 。然而, 即使使用 $k = n$, 可分离卷积的复杂度也等同于一个自注意力层与一个逐位置前馈层的组合, 这正是我们模型所采用的方法。

作为附带好处, 自注意力可以产生更具可解释性的模型。我们检查了模型中的注意力分布, 并在附录中展示和讨论了示例。各个注意力头不仅明确学会执行不同任务, 许多还似乎表现出与句子的句法和语义结构相关的行为。

5 训练

本节介绍我们模型的训练方案。

5.1 训练数据与批处理

我们在标准的 WMT 2014 英德数据集上进行训练, 该数据集包含约 450 万对句子。句子使用字节对编码 [3], 进行编码, 其共享的源-目标词汇表约有 37000 个词元。对于英法任务, 我们使用了规模大得多的 WMT 2014 英法数据集, 该数据集包含 3600 万句, 并将词元切分为包含 32000 个词片的词汇表 [38]。句子对按近似序列长度分批。每个训练批次包含一组句子对, 其中约有 25000 个源词元和 25000 个目标词元。

5.2 硬件与时间安排

我们在配备 8 块 NVIDIA P100 GPU 的单台机器上训练了模型。对于使用本文所述超参数的基础模型, 每个训练步大约耗时 0.4 秒。我们将基础模型总共训练了 100,000 步, 即 12 小时。对于大型模型(见表 3 最后一行), 每步时间为 1.0 秒。大型模型训练了 300,000 步(3.5 天)。

5.3 优化器

我们使用了 Adam 优化器 [20], 其中 $\beta_1 = 0.9$ 、 $\beta_2 = 0.98$ 和 $\epsilon = 10^{-9}$ 。我们在训练过程中根据以下公式改变学习率:

$$lrate = d_{\text{model}}^{-0.5} \cdot \min(step_num^{-0.5}, step_num \cdot warmup_steps^{-1.5}) \quad (3)$$

这对应于在前 $warmup_steps$ 个训练步中线性增加学习率, 之后则按步数平方根的倒数成比例地降低学习率。我们使用了 $warmup_steps = 4000$ 。

5.4 正则化

我们在训练期间采用三种类型的正则化:

Table 2: The Transformer achieves better BLEU scores than previous state-of-the-art models on the English-to-German and English-to-French newstest2014 tests at a fraction of the training cost.

Model	BLEU		Training Cost (FLOPs)	
	EN-DE	EN-FR	EN-DE	EN-FR
ByteNet [18]	23.75			
Deep-Att + PosUnk [39]		39.2		$1.0 \cdot 10^{20}$
GNMT + RL [38]	24.6	39.92	$2.3 \cdot 10^{19}$	$1.4 \cdot 10^{20}$
ConvS2S [9]	25.16	40.46	$9.6 \cdot 10^{18}$	$1.5 \cdot 10^{20}$
MoE [32]	26.03	40.56	$2.0 \cdot 10^{19}$	$1.2 \cdot 10^{20}$
Deep-Att + PosUnk Ensemble [39]		40.4		$8.0 \cdot 10^{20}$
GNMT + RL Ensemble [38]	26.30	41.16	$1.8 \cdot 10^{20}$	$1.1 \cdot 10^{21}$
ConvS2S Ensemble [9]	26.36	41.29	$7.7 \cdot 10^{19}$	$1.2 \cdot 10^{21}$
Transformer (base model)	27.3	38.1	$3.3 \cdot 10^{18}$	
Transformer (big)	28.4	41.8	$2.3 \cdot 10^{19}$	

Residual Dropout We apply dropout [33] to the output of each sub-layer, before it is added to the sub-layer input and normalized. In addition, we apply dropout to the sums of the embeddings and the positional encodings in both the encoder and decoder stacks. For the base model, we use a rate of $P_{drop} = 0.1$.

Label Smoothing During training, we employed label smoothing of value $\epsilon_{ls} = 0.1$ [36]. This hurts perplexity, as the model learns to be more unsure, but improves accuracy and BLEU score.

6 Results

6.1 Machine Translation

On the WMT 2014 English-to-German translation task, the big transformer model (Transformer (big) in Table 2) outperforms the best previously reported models (including ensembles) by more than 2.0 BLEU, establishing a new state-of-the-art BLEU score of 28.4. The configuration of this model is listed in the bottom line of Table 3. Training took 3.5 days on 8 P100 GPUs. Even our base model surpasses all previously published models and ensembles, at a fraction of the training cost of any of the competitive models.

On the WMT 2014 English-to-French translation task, our big model achieves a BLEU score of 41.0, outperforming all of the previously published single models, at less than 1/4 the training cost of the previous state-of-the-art model. The Transformer (big) model trained for English-to-French used dropout rate $P_{drop} = 0.1$, instead of 0.3.

For the base models, we used a single model obtained by averaging the last 5 checkpoints, which were written at 10-minute intervals. For the big models, we averaged the last 20 checkpoints. We used beam search with a beam size of 4 and length penalty $\alpha = 0.6$ [38]. These hyperparameters were chosen after experimentation on the development set. We set the maximum output length during inference to input length + 50, but terminate early when possible [38].

Table 2 summarizes our results and compares our translation quality and training costs to other model architectures from the literature. We estimate the number of floating point operations used to train a model by multiplying the training time, the number of GPUs used, and an estimate of the sustained single-precision floating-point capacity of each GPU ⁵.

6.2 Model Variations

To evaluate the importance of different components of the Transformer, we varied our base model in different ways, measuring the change in performance on English-to-German translation on the

⁵We used values of 2.8, 3.7, 6.0 and 9.5 TFLOPS for K80, K40, M40 and P100, respectively.

表 2: 在英语到德语和英语到法语 newstest2014 测试集上, Transformer 以极小的训练成本取得了比先前最先进模型更好的 BLEU 分数。

模型	BLEU		训练成本 (FLOPs)	
	英-德	英-法	英-德	英-法
ByteNet [18]	23.75			
Deep-Att + PosUnk [39]		39.2		$1.0 \cdot 10^{20}$
GNMT +RL [38]	24.6	39.92	$2.3 \cdot 10^{19}$	$1.4 \cdot 10^{20}$
ConvS2S [9]	25.16	40.46	$9.6 \cdot 10^{18}$	$1.5 \cdot 10^{20}$
MoE [32]	26.03	40.56	$2.0 \cdot 10^{19}$	$1.2 \cdot 10^{20}$
Deep-Att +PosUnk 集成 [39]		40.4		$8.0 \cdot 10^{20}$
GNMT +RL 集成 [38]	26.30	41.16	$1.8 \cdot 10^{20}$	$1.1 \cdot 10^{21}$
ConvS2S 集成 [9]	26.36	41.29	$7.7 \cdot 10^{19}$	$1.2 \cdot 10^{21}$
Transformer (基础模型)	27.3	38.1	$3.3 \cdot 10^{18}$	
Transformer (大型模型)	28.4	41.8	$2.3 \cdot 10^{19}$	

残差 Dropout 我们对每个子层的输出应用 dropout [33], 在将其加到子层输入并进行归一化之前。此外, 我们对编码器和解码器堆栈中的嵌入与位置编码之和也应用 dropout。对于基础模型, 我们使用的 dropout 比率为 $P_{drop} = 0.1$ 。

标签平滑 训练期间, 我们采用了值为 $\epsilon_{ls} = 0.1$ [36]的标签平滑。这会损害困惑度, 因为模型会学会更加不确定, 但会提高准确率和 BLEU 分数。

6 结果

6.1 机器翻译

在 WMT 2014 英语到德语翻译任务上, 大型 Transformer 模型 (表 2 中的 Transformer (big)) 以超过 2.0 BLEU 的优势优于此前报告的最佳模型 (包括集成模型), 创造了 28.4 的 BLEU 新最优成绩。该模型的配置列于表 3 的最后一行。训练在 8 块 P100 GPU 上耗时 3.5 天。甚至我们的基础模型也超越了此前发表的所有模型和集成模型, 而训练成本仅为任何有竞争力的模型的一小部分。

在 WMT 2014 英语到法语翻译任务上, 我们的大型模型取得了 41.0 的 BLEU 分数, 优于此前发表的所有单模型, 且训练成本不到之前最先进模型训练成本的 1/4。用于英语到法语训练的 Transformer (big) 模型使用了 dropout 率 $P_{drop} = 0.1$, 而不是 0.3。

对于基础模型, 我们使用通过平均最后 5 个检查点获得的单个模型, 这些检查点每 10 分钟写入一次。对于大型模型, 我们平均了最后 20 个检查点。我们使用束搜索, 束大小为 4, 长度惩罚为 $\alpha = 0.6$ [38]。这些超参数是在开发集上实验后选择的。我们将推理期间的最大输出长度设置为输入长度 + 50, 但在可能时提前终止 [38]。

表 2 总结了我们的结果, 并将我们的翻译质量和训练成本与文献中的其他模型架构进行了比较。我们通过将训练时间、使用的 GPU 数量以及每个 GPU 的持续单精度浮点运算能力的估计值相乘来估计训练模型所用的浮点运算次数 ⁵。

6.2 模型变体

为了评估 Transformer 中不同组件的重要性, 我们以不同方式改变我们的基础模型, 测量英德翻译性能的变化, 在

⁵我们使用了 2.8、3.7、6.0 和 9.5 TFLOPS, 适用于 K80、K40、M40 和 P100, 分别。

Table 3: Variations on the Transformer architecture. Unlisted values are identical to those of the base model. All metrics are on the English-to-German translation development set, newstest2013. Listed perplexities are per-wordpiece, according to our byte-pair encoding, and should not be compared to per-word perplexities.

	N	d_{model}	d_{ff}	h	d_k	d_v	P_{drop}	ϵ_{ts}	train steps	PPL (dev)	BLEU (dev)	params $\times 10^6$		
base	6	512	2048	8	64	64	0.1	0.1	100K	4.92	25.8	65		
(A)					1	512	512				5.29	24.9		
					4	128	128				5.00	25.5		
					16	32	32				4.91	25.8		
					32	16	16				5.01	25.4		
(B)					16					5.16	25.1	58		
					32					5.01	25.4	60		
(C)	2									6.11	23.7	36		
	4									5.19	25.3	50		
	8									4.88	25.5	80		
		256					32	32			5.75	24.5	28	
		1024					128	128			4.66	26.0	168	
			1024									5.12	25.4	53
			4096									4.75	26.2	90
(D)							0.0			5.77	24.6			
							0.2			4.95	25.5			
								0.0		4.67	25.3			
								0.2		5.47	25.7			
(E)	positional embedding instead of sinusoids									4.92	25.7			
big	6	1024	4096	16					0.3	300K	4.33	26.4	213	

development set, newstest2013. We used beam search as described in the previous section, but no checkpoint averaging. We present these results in Table 3.

In Table 3 rows (A), we vary the number of attention heads and the attention key and value dimensions, keeping the amount of computation constant, as described in Section 3.2.2. While single-head attention is 0.9 BLEU worse than the best setting, quality also drops off with too many heads.

In Table 3 rows (B), we observe that reducing the attention key size d_k hurts model quality. This suggests that determining compatibility is not easy and that a more sophisticated compatibility function than dot product may be beneficial. We further observe in rows (C) and (D) that, as expected, bigger models are better, and dropout is very helpful in avoiding over-fitting. In row (E) we replace our sinusoidal positional encoding with learned positional embeddings [9], and observe nearly identical results to the base model.

6.3 English Constituency Parsing

To evaluate if the Transformer can generalize to other tasks we performed experiments on English constituency parsing. This task presents specific challenges: the output is subject to strong structural constraints and is significantly longer than the input. Furthermore, RNN sequence-to-sequence models have not been able to attain state-of-the-art results in small-data regimes [37].

We trained a 4-layer transformer with $d_{\text{model}} = 1024$ on the Wall Street Journal (WSJ) portion of the Penn Treebank [25], about 40K training sentences. We also trained it in a semi-supervised setting, using the larger high-confidence and BerkleyParser corpora from with approximately 17M sentences [37]. We used a vocabulary of 16K tokens for the WSJ only setting and a vocabulary of 32K tokens for the semi-supervised setting.

We performed only a small number of experiments to select the dropout, both attention and residual (section 5.4), learning rates and beam size on the Section 22 development set, all other parameters remained unchanged from the English-to-German base translation model. During inference, we

表3: Transformer 架构的变体。未列出的值与基础模型相同。所有指标均在英德翻译开发集 newstest2013 上。列出的困惑度是按 wordpiece 计算的，依据我们的字节对编码，不应与按词计算的困惑度相比较。

	N	$d_{\text{模型}}$	d_{ff}	h	d_k	d_v	P_{drop}	ϵ_{ls}	训练 步数	PPL (开发集)	BLEU (开发集)	参数 $\times 10^6$	
base	6	512	2048	8	64	64	0.1	0.1	100K	4.92	25.8	65	
(A)					1	512	512				5.29	24.9	
					4	128	128				5.00	25.5	
					16	32	32				4.91	25.8	
					32	16	16				5.01	25.4	
(B)					16						5.16	25.1	58
					32						5.01	25.4	60
(C)	2									6.11	23.7	36	
	4									5.19	25.3	50	
	8									4.88	25.5	80	
		256			32	32				5.75	24.5	28	
		1024			128	128				4.66	26.0	168	
			1024							5.12	25.4	53	
			4096							4.75	26.2	90	
(D)							0.0			5.77	24.6		
							0.2			4.95	25.5		
								0.0		4.67	25.3		
								0.2		5.47	25.7		
(E)	用位置嵌入代替正弦函数									4.92	25.7		
big	6	1024	4096	16				0.3	300K	4.33	26.4	213	

开发集 newstest2013。我们使用上一节所述的束搜索，但没有使用检查点平均。我们在表 3 中展示这些结果。

在表 3 的行 (A) 中，我们改变注意力头的数量以及注意力键和值的维度，同时保持计算量不变，如第 3.2.2 节所述。虽然单头注意力比最佳设置差 0.9 BLEU，但当头数过多时，质量也会下降。

在表 3 的行 (B) 中，我们观察到减小注意力键的大小 d_k 会损害模型质量。这表明，确定兼容性并非易事，而采用比点积更复杂的兼容性函数可能是有益的。我们在行 (C) 和 (D) 中进一步观察到，正如预期，更大的模型效果更好，而 dropout 对于避免过拟合非常有帮助。在行 (E) 中，我们用学习到的位置嵌入 [9].替换正弦位置编码，并观察到与基础模型几乎相同的结果。

6.3 英语成分句法分析

为了评估 Transformer 是否能泛化到其他任务，我们在英语成分句法分析上进行了实验。该任务带来了特定挑战：输出受到强结构约束，并且明显长于输入。此外，RNN 序列到序列模型一直无法在小数据场景下取得最先进的结果 [37]。

我们使用 $d_{\text{model}} = 1024$ 在 Penn Treebank [25], 的 Wall Street Journal (WSJ) 部分上，用约 4 万条训练句子训练了一个 4 层 Transformer。我们还在半监督设置下对其进行了训练，使用了更大的高置信度和 BerkleyParser 语料库，其中约 1700 万句[37]。我们为仅 WSJ 设置使用了包含 16K 个 token 的词表，并为半监督设置使用了包含 32K 个 token 的词表。

我们在第22节开发集上仅对 dropout（包括注意力和残差，见第5.4节）、学习率和束大小进行了少量实验，其他所有参数均与英德基础翻译模型保持一致。在推理时，我们

Table 4: The Transformer generalizes well to English constituency parsing (Results are on Section 23 of WSJ)

Parser	Training	WSJ 23 F1
Vinyals & Kaiser el al. (2014) [37]	WSJ only, discriminative	88.3
Petrov et al. (2006) [29]	WSJ only, discriminative	90.4
Zhu et al. (2013) [40]	WSJ only, discriminative	90.4
Dyer et al. (2016) [8]	WSJ only, discriminative	91.7
Transformer (4 layers)	WSJ only, discriminative	91.3
Zhu et al. (2013) [40]	semi-supervised	91.3
Huang & Harper (2009) [14]	semi-supervised	91.3
McClosky et al. (2006) [26]	semi-supervised	92.1
Vinyals & Kaiser el al. (2014) [37]	semi-supervised	92.1
Transformer (4 layers)	semi-supervised	92.7
Luong et al. (2015) [23]	multi-task	93.0
Dyer et al. (2016) [8]	generative	93.3

increased the maximum output length to input length + 300. We used a beam size of 21 and $\alpha = 0.3$ for both WSJ only and the semi-supervised setting.

Our results in Table 4 show that despite the lack of task-specific tuning our model performs surprisingly well, yielding better results than all previously reported models with the exception of the Recurrent Neural Network Grammar [8].

In contrast to RNN sequence-to-sequence models [37], the Transformer outperforms the Berkeley-Parser [29] even when training only on the WSJ training set of 40K sentences.

7 Conclusion

In this work, we presented the Transformer, the first sequence transduction model based entirely on attention, replacing the recurrent layers most commonly used in encoder-decoder architectures with multi-headed self-attention.

For translation tasks, the Transformer can be trained significantly faster than architectures based on recurrent or convolutional layers. On both WMT 2014 English-to-German and WMT 2014 English-to-French translation tasks, we achieve a new state of the art. In the former task our best model outperforms even all previously reported ensembles.

We are excited about the future of attention-based models and plan to apply them to other tasks. We plan to extend the Transformer to problems involving input and output modalities other than text and to investigate local, restricted attention mechanisms to efficiently handle large inputs and outputs such as images, audio and video. Making generation less sequential is another research goals of ours.

The code we used to train and evaluate our models is available at <https://github.com/tensorflow/tensor2tensor>.

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References

[1] Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E Hinton. Layer normalization. *arXiv preprint arXiv:1607.06450*, 2016.

[2] Dzmitry Bahdanau, Kyunghyun Cho, and Yoshua Bengio. Neural machine translation by jointly learning to align and translate. *CoRR*, abs/1409.0473, 2014.

[3] Denny Britz, Anna Goldie, Minh-Thang Luong, and Quoc V. Le. Massive exploration of neural machine translation architectures. *CoRR*, abs/1703.03906, 2017.

[4] Jianpeng Cheng, Li Dong, and Mirella Lapata. Long short-term memory-networks for machine reading. *arXiv preprint arXiv:1601.06733*, 2016.

表 4: Transformer 能很好地泛化到英语成分句法分析（结果基于 WSJ 第 23 节）

解析器	训练	WSJ 23 F1
Vinyals & Kaiser 等人 (2014) [37]	仅 WSJ, 判别式	88.3
Petrov 等 (2006) [29]	仅限 WSJ, 判别式	90.4
Zhu 等 (2013) [40]	仅限 WSJ, 判别式	90.4
Dyer 等 (2016) [8]	仅限 WSJ, 判别式	91.7
Transformer (4 层)	仅 WSJ, 判别式	91.3
Zhu 等人 (2013) [40]	半监督	91.3
Huang 和 Harper (2009) [14]	半监督	91.3
McClosky 等人 (2006) [26]	半监督	92.1
Vinyals & Kaiser 等人 (2014) [37]	半监督	92.1
Transformer (4 层)	半监督	92.7
Luong 等人 (2015) [23]	多任务	93.0
Dyer 等人 (2016) [8]	生成式	93.3

incr将最大输出长度放宽至输入长度 + 300。我们使用了大小为 21 的束搜索，以及 $\alpha = 0.3$ for 无论是仅使用 WSJ 还是半监督设置。

我们在表 4 中的结果表明，尽管缺乏针对特定任务的调优，我们的模型仍表现得出奇地好，取得了优于所有此前报告的模型的结果，但循环神经网络语法 [8] 除外。

与 RNN 序列到序列模型相比 [37],, Transformer 即使仅在 40K 句子的 WSJ 训练集上训练，也优于 Berkeley-Parser [29]。

7 结论

在这项工作中，我们提出了 Transformer，这是第一个完全基于注意力的序列转导模型；它取代了编码器-解码器架构中最常用的循环层，改用多头自注意力。

对于翻译任务，Transformer 的训练速度可以显著快于基于循环层或卷积层的架构。在 WMT 2014 英德和 WMT 2014英法翻译任务上，我们均取得了新的最优结果。在前一项任务中，我们最好的模型甚至优于此前报告的所有集成模型。

我们对基于注意力的模型的未来感到兴奋，并计划将其应用于其他任务。我们计划将 Transformer 扩展到涉及文本以外输入和输出模态的问题，并研究局部、受限的注意力机制，以高效处理大规模输入和输出，例如图像、音频和视频。让生成过程减少顺序依赖也是我们另一个研究目标。

我们用于训练和评估模型的代码可在 <https://github.com/tensorflow/tensor2tensor> 获取。

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参考文献

[1] Jimmy Lei Ba、Jamie Ryan Kiros 和 Geoffrey E Hinton。层归一化。arXiv 预印本arXiv:1607.06450, 2016。[2] Dzmitry Bahdanau、Kyunghyun Cho 和 Yoshua Bengio。通过联合学习对齐和翻译的神经机器翻译。*CoRR*, abs/1409.0473, 2014。[3] Denny Britz、Anna Goldie、Minh-Thang Luong 和 Quoc V. Le。神经机器翻译架构的大规模探索。*CoRR*, abs/1703.03906, 2017。[4] Jianpeng Cheng、Li Dong 和 Mirella Lapata。用于机器阅读的长短期记忆网络。*arXiv* 预印本 *arXiv:1601.06733*, 2016。

- [5] Kyunghyun Cho, Bart van Merriënboer, Çağlar Gülçehre, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. Learning phrase representations using rnn encoder-decoder for statistical machine translation. *CoRR*, abs/1406.1078, 2014.
- [6] François Chollet. Xception: Deep learning with depthwise separable convolutions. *arXiv preprint arXiv:1610.02357*, 2016.
- [7] Junyoung Chung, Çağlar Gülçehre, Kyunghyun Cho, and Yoshua Bengio. Empirical evaluation of gated recurrent neural networks on sequence modeling. *CoRR*, abs/1412.3555, 2014.
- [8] Chris Dyer, Adhiguna Kuncoro, Miguel Ballesteros, and Noah A. Smith. Recurrent neural network grammars. In *Proc. of NAACL*, 2016.
- [9] Jonas Gehring, Michael Auli, David Grangier, Denis Yarats, and Yann N. Dauphin. Convolutional sequence to sequence learning. *arXiv preprint arXiv:1705.03122v2*, 2017.
- [10] Alex Graves. Generating sequences with recurrent neural networks. *arXiv preprint arXiv:1308.0850*, 2013.
- [11] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 770–778, 2016.
- [12] Sepp Hochreiter, Yoshua Bengio, Paolo Frasconi, and Jürgen Schmidhuber. Gradient flow in recurrent nets: the difficulty of learning long-term dependencies, 2001.
- [13] Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural computation*, 9(8):1735–1780, 1997.
- [14] Zhongqiang Huang and Mary Harper. Self-training PCFG grammars with latent annotations across languages. In *Proceedings of the 2009 Conference on Empirical Methods in Natural Language Processing*, pages 832–841. ACL, August 2009.
- [15] Rafal Jozefowicz, Oriol Vinyals, Mike Schuster, Noam Shazeer, and Yonghui Wu. Exploring the limits of language modeling. *arXiv preprint arXiv:1602.02410*, 2016.
- [16] Łukasz Kaiser and Samy Bengio. Can active memory replace attention? In *Advances in Neural Information Processing Systems, (NIPS)*, 2016.
- [17] Łukasz Kaiser and Ilya Sutskever. Neural GPUs learn algorithms. In *International Conference on Learning Representations (ICLR)*, 2016.
- [18] Nal Kalchbrenner, Lasse Espeholt, Karen Simonyan, Aaron van den Oord, Alex Graves, and Koray Kavukcuoglu. Neural machine translation in linear time. *arXiv preprint arXiv:1610.10099v2*, 2017.
- [19] Yoon Kim, Carl Denton, Luong Hoang, and Alexander M. Rush. Structured attention networks. In *International Conference on Learning Representations*, 2017.
- [20] Diederik Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *ICLR*, 2015.
- [21] Oleksii Kuchaiev and Boris Ginsburg. Factorization tricks for LSTM networks. *arXiv preprint arXiv:1703.10722*, 2017.
- [22] Zhouhan Lin, Minwei Feng, Cicero Nogueira dos Santos, Mo Yu, Bing Xiang, Bowen Zhou, and Yoshua Bengio. A structured self-attentive sentence embedding. *arXiv preprint arXiv:1703.03130*, 2017.
- [23] Minh-Thang Luong, Quoc V. Le, Ilya Sutskever, Oriol Vinyals, and Łukasz Kaiser. Multi-task sequence to sequence learning. *arXiv preprint arXiv:1511.06114*, 2015.
- [24] Minh-Thang Luong, Hieu Pham, and Christopher D Manning. Effective approaches to attention-based neural machine translation. *arXiv preprint arXiv:1508.04025*, 2015.

[5] Kyunghyun Cho、Bart van Merriënboer、Çağlar Gülçehre、Fethi Bougares、Holger Schwenk 和 Yoshua Bengio。使用 RNN 编码器-解码器学习短语表示以进行统计机器翻译。*CoRR*, abs/1406.1078, 2014。[6] François Chollet。Xception：使用深度可分离卷积的深度学习。arXiv预印本 arXiv:1610.02357, 2016。[7] Junyoung Chung、Çağlar Gülçehre、Kyunghyun Cho 和 Yoshua Bengio。门控循环神经网络在序列建模上的实证评估。*CoRR*, abs/1412.3555, 2014。[8] Chris Dyer、Adhiguna Kuncoro、Miguel Ballesteros 和 Noah A. Smith。循环神经网络语法。载于 *NAACL* 会议论文集, 2016。[9] Jonas Gehring、Michael Auli、David Grangier、Denis Yarats 和 Yann N. Dauphin。卷积序列到序列学习。*arXiv* 预印本 *arXiv:1705.03122v2*, 2017。[10] Alex Graves。使用循环神经网络生成序列。arXiv 预印本 arXiv:1308.0850, 2013。[11] Kaiming He、Xiangyu Zhang、Shaoqing Ren 和 Jian Sun。用于图像识别的深度残差学习。载于 *IEEE* 计算机视觉与模式识别会议论文集, 第770–778页, 2016。[12] Sepp Hochreiter、Yoshua Bengio、Paolo Frasconi 和 Jürgen Schmidhuber。循环网络中的梯度流：学习长期依赖的困难, 2001。[13] Sepp Hochreiter 和 Jürgen Schmidhuber。长短期记忆。神经计算, 9(8):1735–1780, 1997。[14] Zhongqiang Huang 和 Mary Harper。跨语言的带有潜在标注的自训练 PCFG 文法。载于 *2009 年自然语言处理经验方法会议论文集*, 第832–841页。ACL, 2009年8月。[15] Rafal Jozefowicz、Oriol Vinyals、Mike Schuster、Noam Shazeer 和 Yonghui Wu。探索语言建模的极限。*arXiv* 预印本 *arXiv:1602.02410*, 2016。[16] Łukasz Kaiser 和 Samy Bengio。主动记忆能取代注意力吗？载于神经信息处理系统进展 (*NIPS*) , 2016。[17] Łukasz Kaiser 和 Ilya Sutskever。神经 GPU 学习算法。载于国际学习表征会议 (*ICLR*) , 2016。[18] Nal Kalchbrenner、Lasse Espeholt、Karen Simonyan、Aaron van den Oord、Alex Graves 和 Koray Kavukcuoglu。线性时间的神经机器翻译。*arXiv* 预印本 *arXiv:1610.10099v2*, 2017。[19] Yoon Kim、Carl Denton、Luong Hoang 和 Alexander M. Rush。结构化注意力网络。载于国际学习表征会议, 2017。[20] Diederik Kingma 和 Jimmy Ba。Adam：一种随机优化方法。载于 *ICLR*, 2015。[21] Oleksii Kuchaiev 和 Boris Ginsburg。LSTM 网络的分解技巧。arXiv 预印本 arXiv:1703.10722, 2017。[22] Zhouhan Lin、Minwei Feng、Cicero Nogueira dos Santos、Mo Yu、Bing Xiang、Bowen Zhou 和 Yoshua Bengio。一种结构化的自注意力句子嵌入。arXiv 预印本 arXiv:1703.03130, 2017。[23] Minh-Thang Luong、Quoc V. Le、Ilya Sutskever、Oriol Vinyals 和 Łukasz Kaiser。多任务序列到序列学习。*arXiv* 预印本 *arXiv:1511.06114*, 2015。[24] Minh-Thang Luong、Hieu Pham 和 Christopher D Manning。基于注意力的神经机器翻译的有效方法。*arXiv* 预印本 *arXiv:1508.04025*, 2015。

- [25] Mitchell P Marcus, Mary Ann Marcinkiewicz, and Beatrice Santorini. Building a large annotated corpus of english: The penn treebank. *Computational linguistics*, 19(2):313–330, 1993.
- [26] David McClosky, Eugene Charniak, and Mark Johnson. Effective self-training for parsing. In *Proceedings of the Human Language Technology Conference of the NAACL, Main Conference*, pages 152–159. ACL, June 2006.
- [27] Ankur Parikh, Oscar Täckström, Dipanjan Das, and Jakob Uszkoreit. A decomposable attention model. In *Empirical Methods in Natural Language Processing*, 2016.
- [28] Romain Paulus, Caiming Xiong, and Richard Socher. A deep reinforced model for abstractive summarization. *arXiv preprint arXiv:1705.04304*, 2017.
- [29] Slav Petrov, Leon Barrett, Romain Thibaux, and Dan Klein. Learning accurate, compact, and interpretable tree annotation. In *Proceedings of the 21st International Conference on Computational Linguistics and 44th Annual Meeting of the ACL*, pages 433–440. ACL, July 2006.
- [30] Ofir Press and Lior Wolf. Using the output embedding to improve language models. *arXiv preprint arXiv:1608.05859*, 2016.
- [31] Rico Sennrich, Barry Haddow, and Alexandra Birch. Neural machine translation of rare words with subword units. *arXiv preprint arXiv:1508.07909*, 2015.
- [32] Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarsz, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. *arXiv preprint arXiv:1701.06538*, 2017.
- [33] Nitish Srivastava, Geoffrey E Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov. Dropout: a simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research*, 15(1):1929–1958, 2014.
- [34] Sainbayar Sukhbaatar, Arthur Szlam, Jason Weston, and Rob Fergus. End-to-end memory networks. In C. Cortes, N. D. Lawrence, D. D. Lee, M. Sugiyama, and R. Garnett, editors, *Advances in Neural Information Processing Systems* 28, pages 2440–2448. Curran Associates, Inc., 2015.
- [35] Ilya Sutskever, Oriol Vinyals, and Quoc VV Le. Sequence to sequence learning with neural networks. In *Advances in Neural Information Processing Systems*, pages 3104–3112, 2014.
- [36] Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jonathon Shlens, and Zbigniew Wojna. Rethinking the inception architecture for computer vision. *CoRR*, abs/1512.00567, 2015.
- [37] Vinyals & Kaiser, Koo, Petrov, Sutskever, and Hinton. Grammar as a foreign language. In *Advances in Neural Information Processing Systems*, 2015.
- [38] Yonghui Wu, Mike Schuster, Zhifeng Chen, Quoc V Le, Mohammad Norouzi, Wolfgang Macherey, Maxim Krikun, Yuan Cao, Qin Gao, Klaus Macherey, et al. Google’s neural machine translation system: Bridging the gap between human and machine translation. *arXiv preprint arXiv:1609.08144*, 2016.
- [39] Jie Zhou, Ying Cao, Xuguang Wang, Peng Li, and Wei Xu. Deep recurrent models with fast-forward connections for neural machine translation. *CoRR*, abs/1606.04199, 2016.
- [40] Muhua Zhu, Yue Zhang, Wenliang Chen, Min Zhang, and Jingbo Zhu. Fast and accurate shift-reduce constituent parsing. In *Proceedings of the 51st Annual Meeting of the ACL (Volume 1: Long Papers)*, pages 434–443. ACL, August 2013.

- [25] Mitchell P Marcus、Mary Ann Marcinkiewicz 和 Beatrice Santorini。构建大型英语标注语料库：宾州树库。计算语言学，19(2):313–330，1993。[26] David McClosky、Eugene Charniak 和 Mark Johnson。用于句法分析的有效自训练。载于 NAACL 人类语言技术会议主会议论文集，第152–159页。ACL，2006年6月。[27] Ankur Parikh、Oscar Täckström、Dipanjan Das 和 Jakob Uszkoreit。一种可分解的注意力模型。载于自然语言处理经验方法，2016。[28] Romain Paulus、Caiming Xiong 和 Richard Socher。用于抽象式摘要的深度强化模型。*arXiv* 预印本 *arXiv:1705.04304*，2017。[29] Slav Petrov、Leon Barrett、Romain Thibaux 和 Dan Klein。学习准确、紧凑且可解释的树标注。载于 第21届国际计算语言学会议暨第44届ACL年会论文集，第433–440页。ACL，2006年7月。[30] Ofir Press 和 Lior Wolf。利用输出嵌入改进语言模型。arXiv预印本 arXiv:1608.05859，2016。[31] Rico Sennrich、Barry Haddow 和 Alexandra Birch。使用子词单元进行稀有词的神经机器翻译。*arXiv* 预印本 *arXiv:1508.07909*，2015。[32] Noam Shazeer、Azalia Mirhoseini、Krzysztof Maziarsz、Andy Davis、Quoc Le、Geoffrey Hinton 和 Jeff Dean。超大规模神经网络：稀疏门控专家混合层。*arXiv* 预印本 *arXiv:1701.06538*，2017。[33] Nitish Srivastava、Geoffrey E Hinton、Alex Krizhevsky、Ilya Sutskever 和 Ruslan Salakhutdinov。Dropout：一种防止神经网络过拟合的简单方法。机器学习研究杂志，15(1):1929–1958，2014。[34] Sainbayar Sukhbaatar、Arthur Szlam、Jason Weston 和 Rob Fergus。端到端记忆网络。载于 C. Cortes、N. D. Lawrence、D. D. Lee、M. Sugiyama 和 R. Garnett 编，神经信息处理系统进展 28，第2440–2448页。Curran Associates 公司，2015。[35] Ilya Sutskever、Oriol Vinyals 和 Quoc VV Le。使用神经网络进行序列到序列学习。载于 神经信息处理系统进展，第3104–3112页，2014。[36] Christian Szegedy、Vincent Vanhoucke、Sergey Ioffe、Jonathon Shlens 和 Zbigniew Wojna。重新思考用于计算机视觉的 Inception 架构。*CoRR*，abs/1512.00567，2015。[37] Vinyals 与 Kaiser、Koo、Petrov、Sutskever 和 Hinton。语法作为外语。载于神经信息处理系统进展，2015。[38] Yonghui Wu、Mike Schuster、Zhifeng Chen、Quoc V Le、Mohammad Norouzi、Wolfgang Macherey、Maxim Krikun、Yuan Cao、Qin Gao、Klaus Macherey 等。谷歌神经机器翻译系统：弥合人类翻译与机器翻译之间的差距。arXiv 预印本arXiv:1609.08144，2016。[39] Jie Zhou、Ying Cao、Xuguang Wang、Peng Li 和 Wei Xu。用于神经机器翻译的带快速前向连接的深度循环模型。*CoRR*，abs/1606.04199，2016。[40] Muhua Zhu、Yue Zhang、Wenliang Chen、Min Zhang 和 Jingbo Zhu。快速准确的移进-归约成分句法分析。载于 第51届ACL年会论文集（第1卷：长论文），第434–443页。ACL，2013年8月。

Attention Visualizations

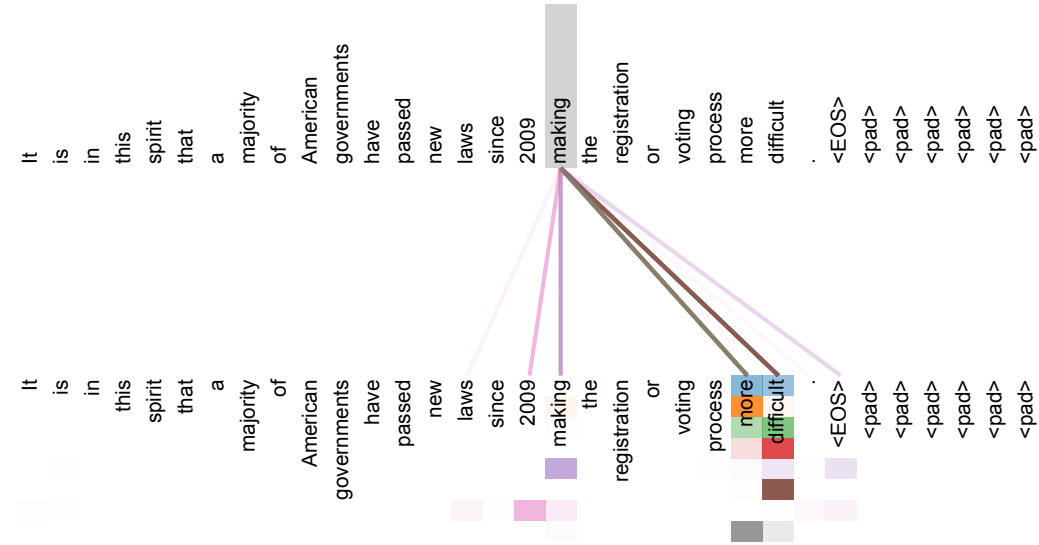


Figure 3: An example of the attention mechanism following long-distance dependencies in the encoder self-attention in layer 5 of 6. Many of the attention heads attend to a distant dependency of the verb ‘making’, completing the phrase ‘making...more difficult’. Attentions here shown only for the word ‘making’. Different colors represent different heads. Best viewed in color.

注意力可视化

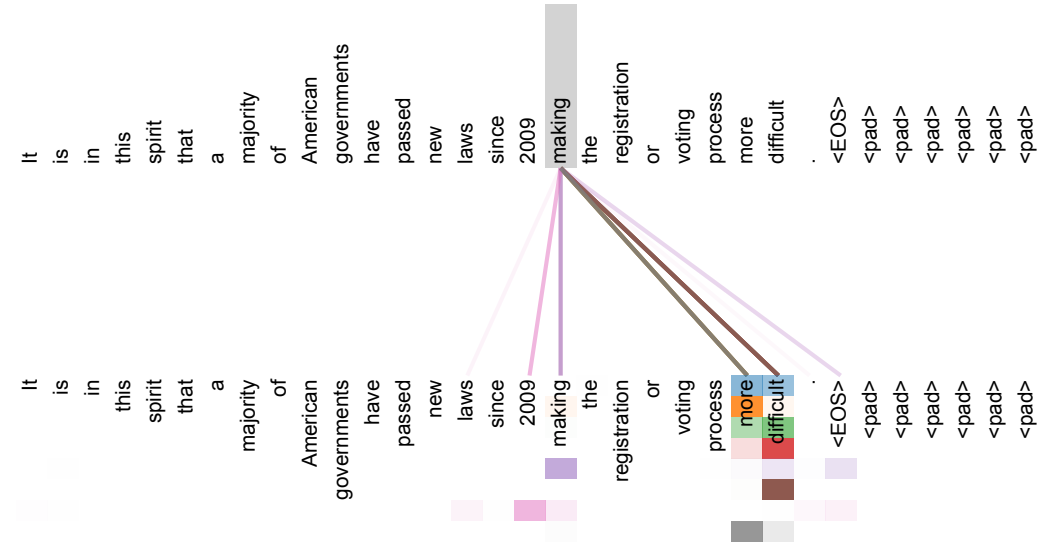


图3：在6层编码器自注意力的第5层中，注意力机制跟随长距离依赖的一个示例。许多注意力头关注动词“making”的一个远距离依赖，补全短语“making...more difficult”。此处仅展示单词“making”的注意力。不同颜色代表不同的注意力头。最好以彩色查看。

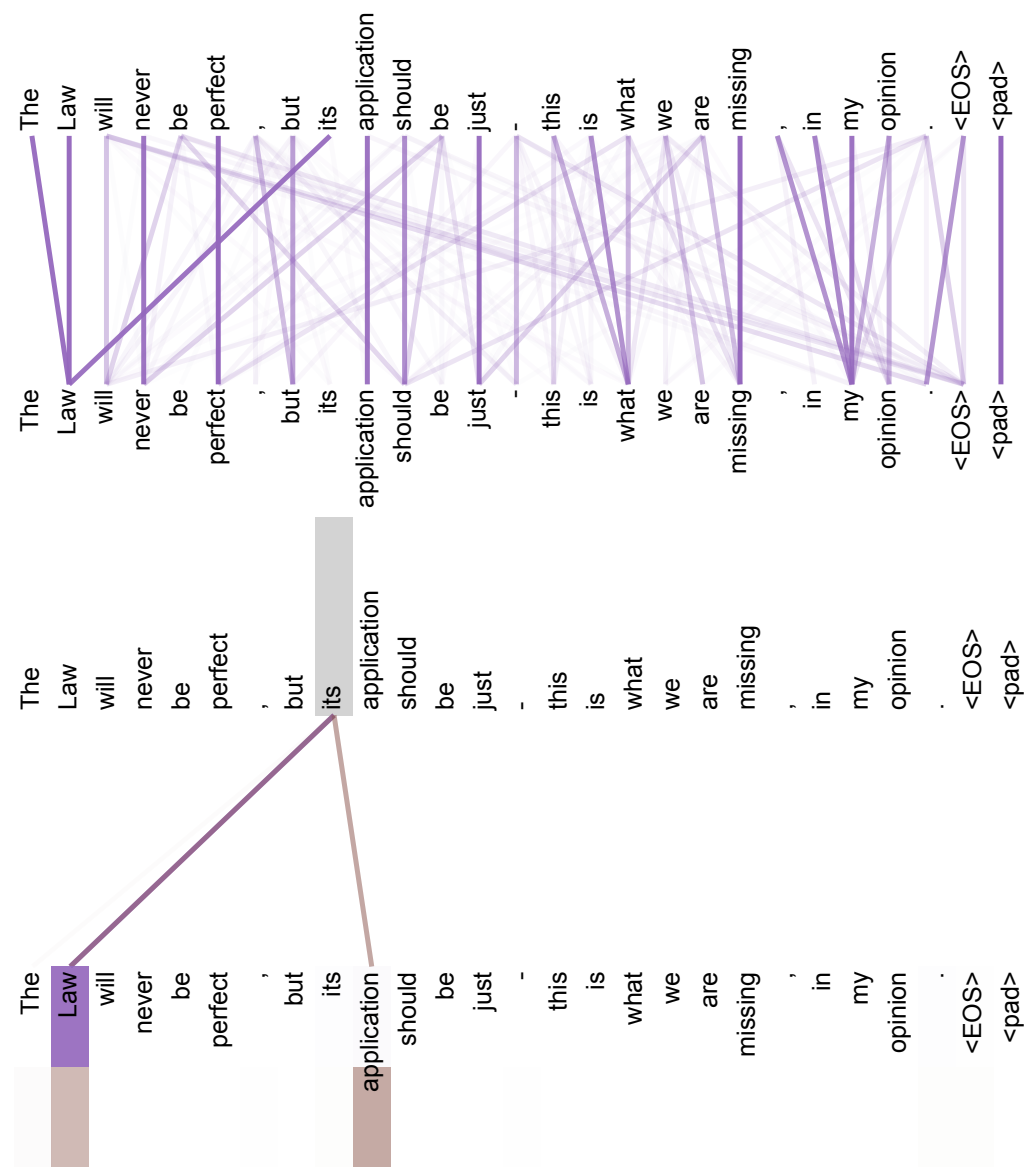


Figure 4: Two attention heads, also in layer 5 of 6, apparently involved in anaphora resolution. Top: Full attentions for head 5. Bottom: Isolated attentions from just the word ‘its’ for attention heads 5 and 6. Note that the attentions are very sharp for this word.

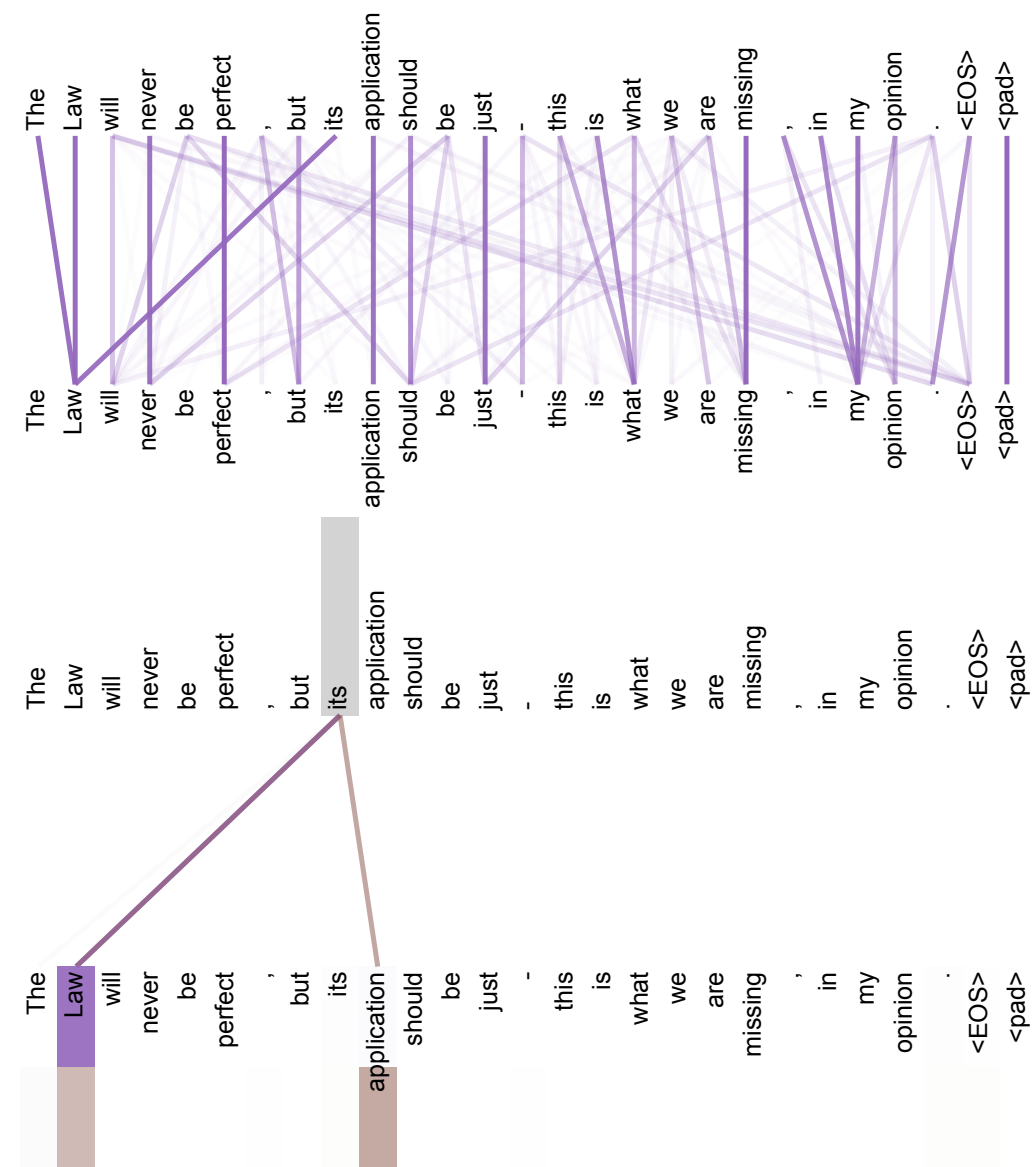


图4：两个注意力头，同样位于6层中的第5层，显然参与了指代消解。上：注意力头5的完整注意力。下：仅来自单词‘its’的、针对注意力头5和6的孤立注意力。请注意，这些注意力对于该词非常集中。

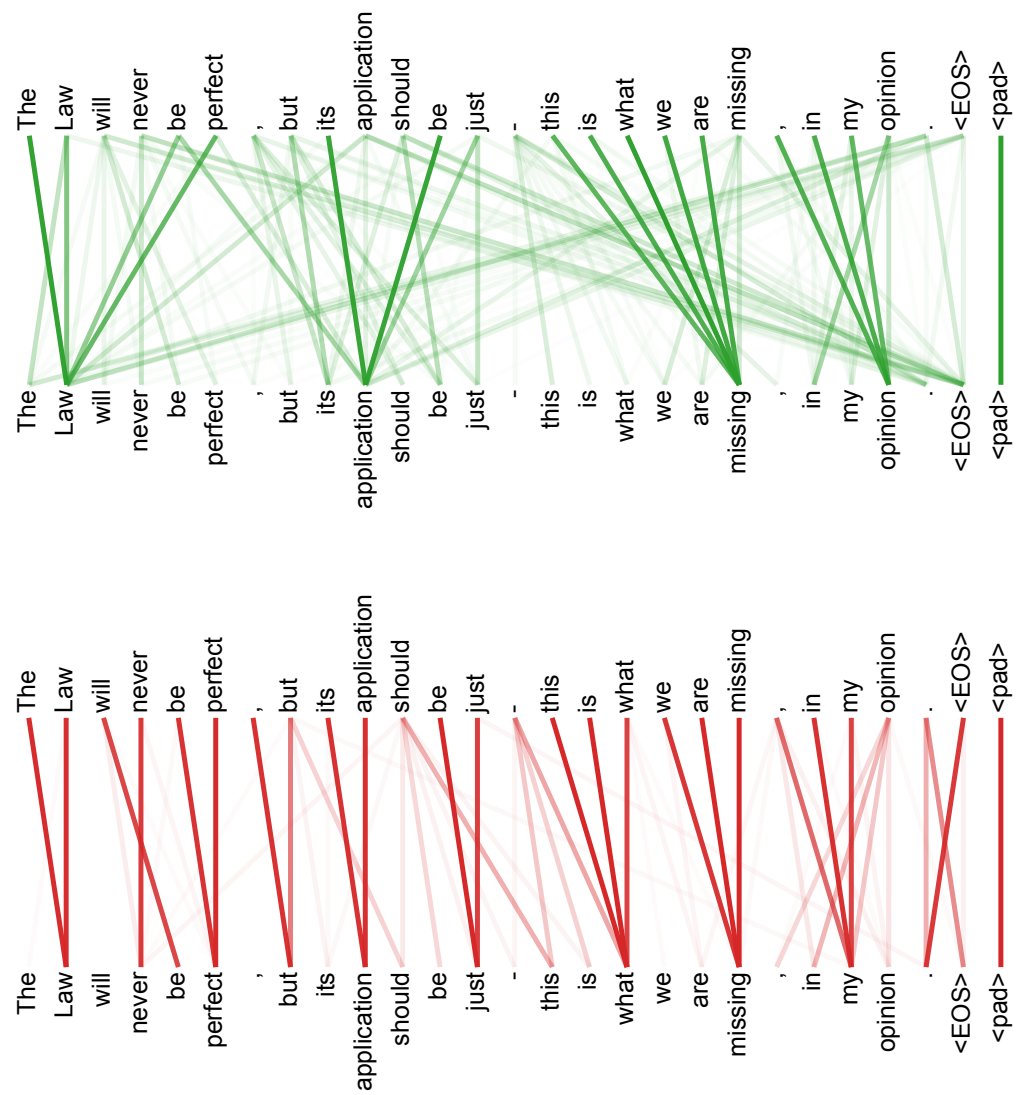


Figure 5: Many of the attention heads exhibit behaviour that seems related to the structure of the sentence. We give two such examples above, from two different heads from the encoder self-attention at layer 5 of 6. The heads clearly learned to perform different tasks.

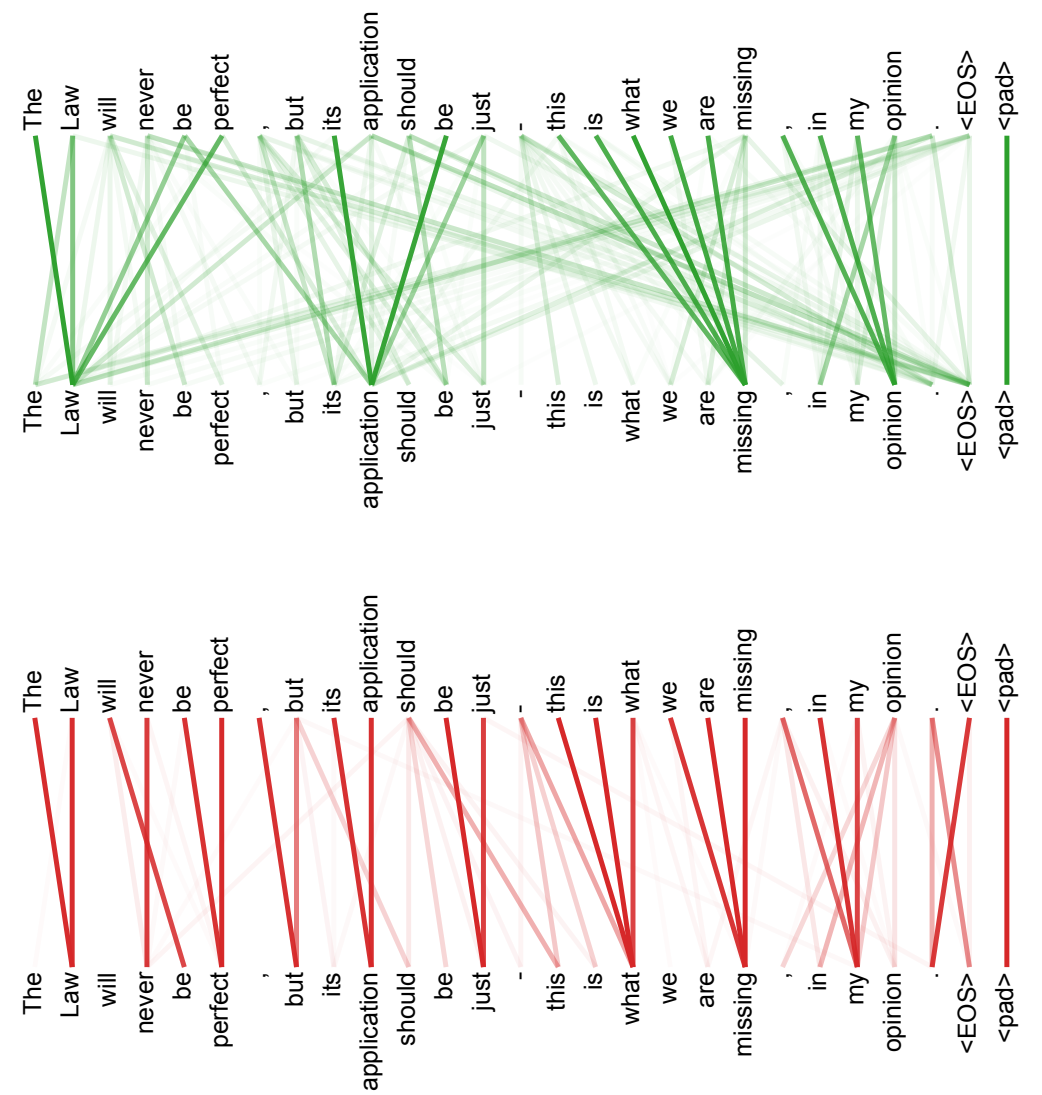


图5：许多注意力头表现出似乎与句子结构相关的行为。我们在上方给出了两个这样的例子，它们来自6层中第5层编码器自注意力的两个不同注意力头。这些注意力头显然学会了执行不同的任务。